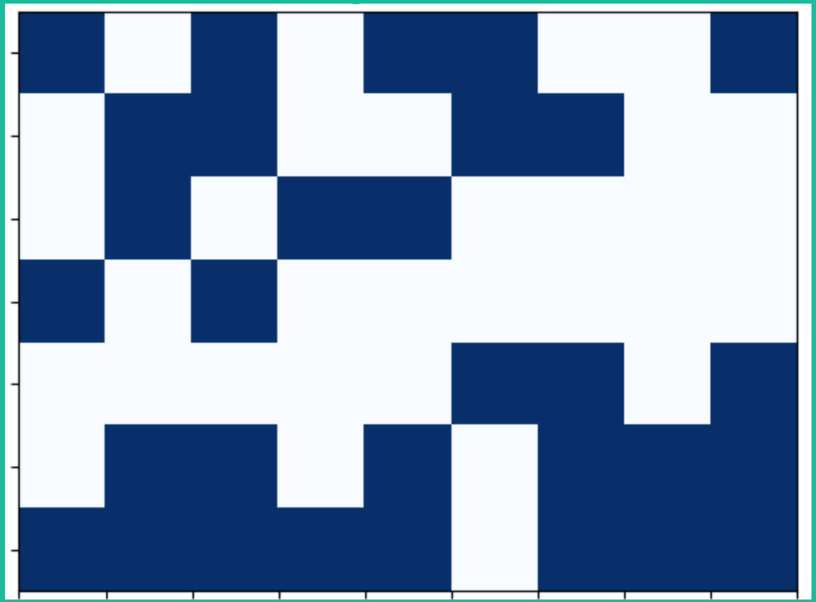


Final Assignment: Economic Complexity Report

Group 4

PART 1: Descriptive Statistics (1)

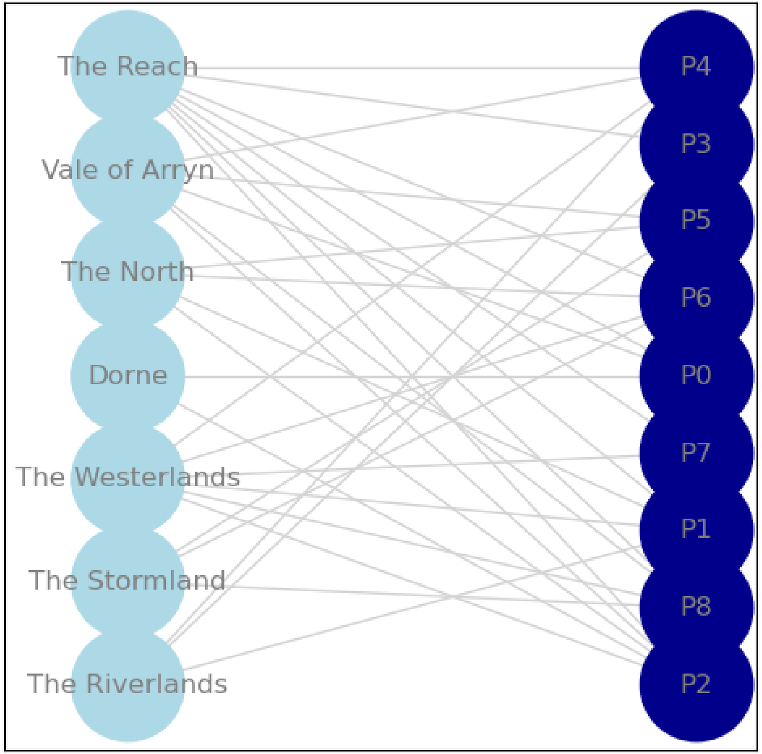
- Let’s start with the input data

```python
# Define matrix, connecting (imaginary) countries and activities.

countries=['Vale of Arryn','The North','The Riverlands', 'Dorne', 'The Stormland','The Westerlands','The Reach']
activities=[f'P{j}' for j in range(9)]

matrix = [[1, 0, 1, 0, 1, 1, 0, 0, 1],
          [0, 1, 1, 0, 0, 1, 1, 0, 0],
          [0, 1, 0, 1, 1, 0, 0, 0, 0],
          [1, 0, 1, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 0, 0, 1, 1, 0, 1],
          [0, 1, 1, 0, 1, 0, 1, 1, 1],
          [1, 1, 1, 1, 1, 0, 1, 1, 1]]
```

PART 1: Descriptive Statistics (2)

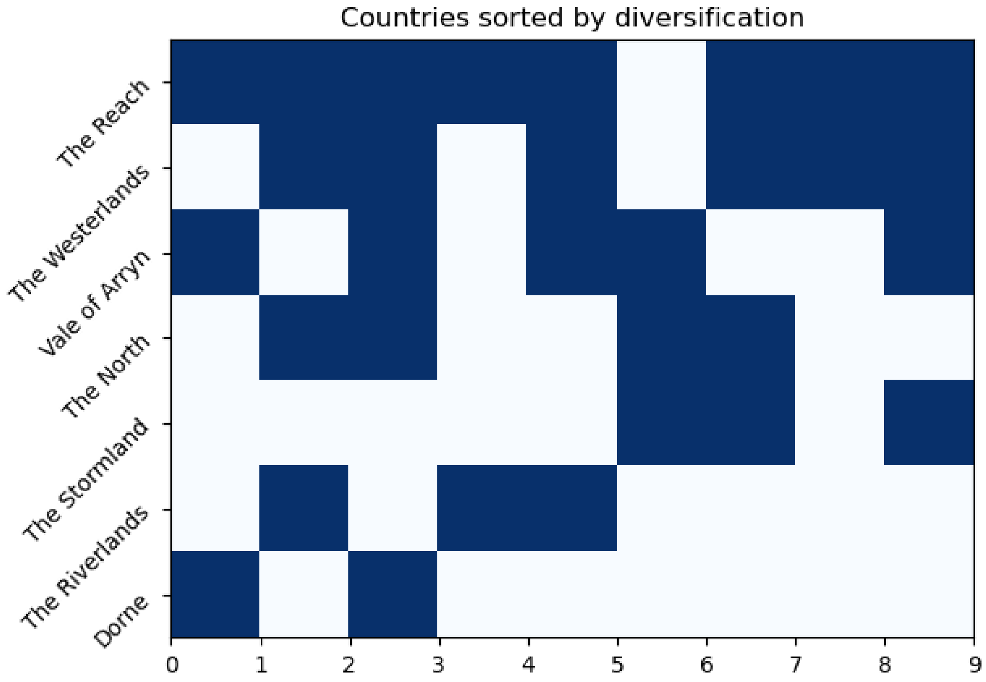
```
Original countries:
        index ' 0' - Vale of Arryn
        index ' 1' - The North
        index ' 2' - The Riverlands
        index ' 3' - Dorne
        index ' 4' - The Stormland
        index ' 5' - The Westerlands
        index ' 6' - The Reach
List of activities:
        index ' 0' - P0
        index ' 1' - P1
        index ' 2' - P2
        index ' 3' - P3
        index ' 4' - P4
        index ' 5' - P5
        index ' 6' - P6
        index ' 7' - P7
        index ' 8' - P8
```

Directed graph



PART 1: Descriptive Statistics (3)

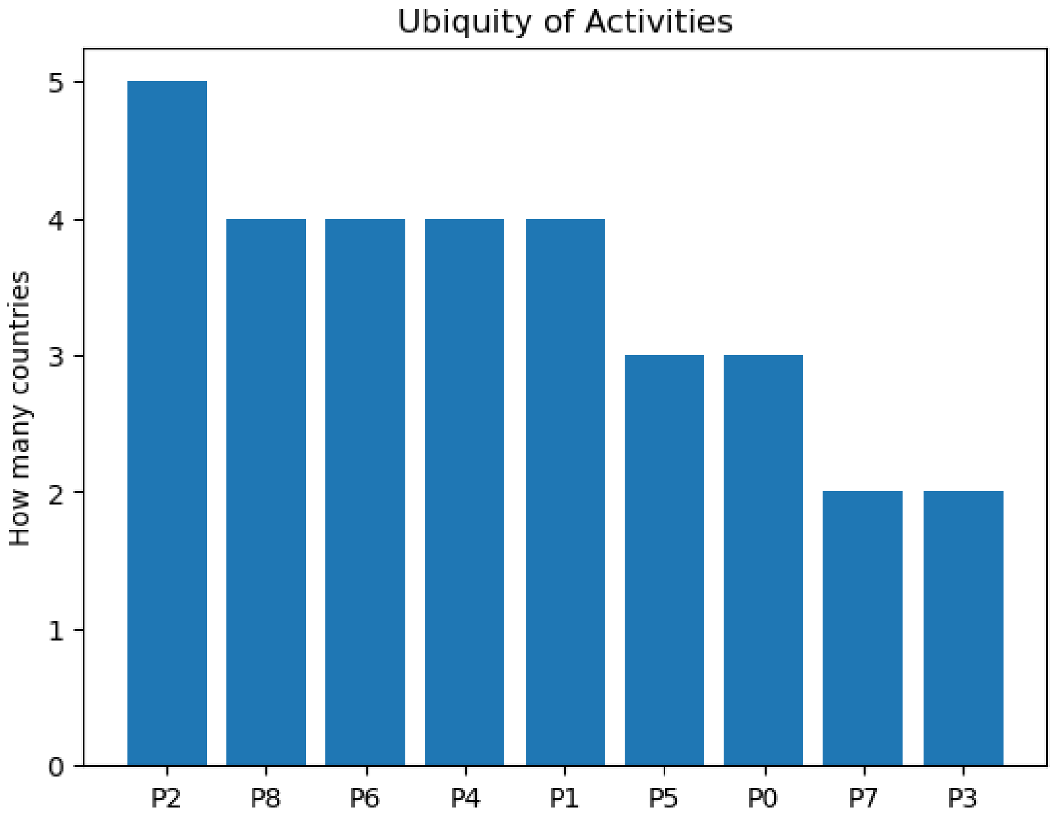
```
Countries sorted by diversification (Most diverse at the top):
        1. - The Reach              (diversification: 8)
        2. - The Westerlands        (diversification: 6)
        3. - Vale of Arryn          (diversification: 5)
        4. - The North              (diversification: 4)
        5. - The Stormland          (diversification: 3)
        6. - The Riverlands         (diversification: 3)
        7. - Dorne                  (diversification: 2)
--- --- ---
```



PART 1: Descriptive Statistics (4)

- **Ubiquity of Products / Activities ("Specialization"):**

  The most common product is **P2** as five countries (out of 7) are able to produce it; while **P7** and **P3** are the most complex to produce as only two countries are able to do so



PART 2: Economic Complexity & Fitness Measures (1)

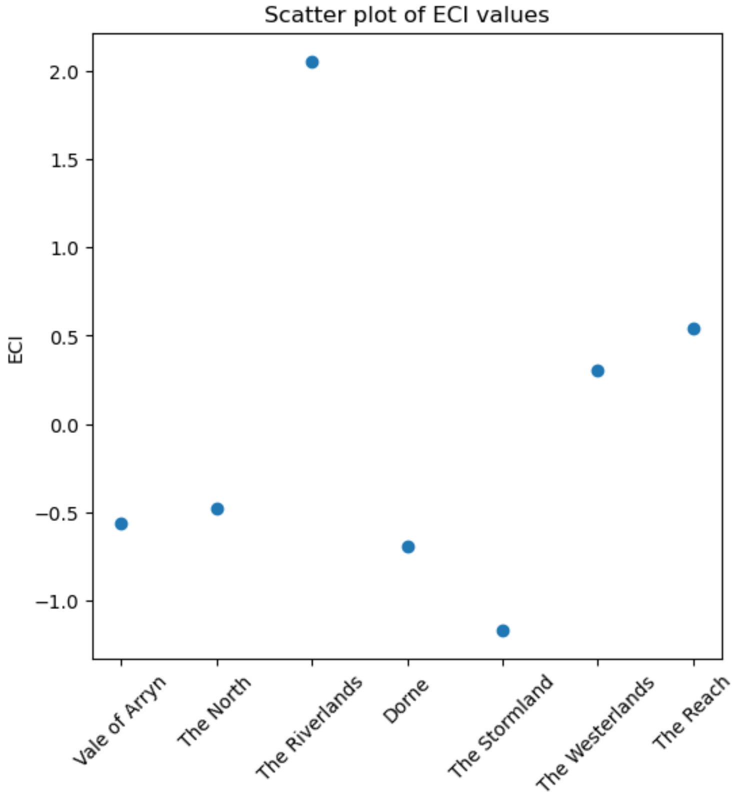
- ## ECI with eigenvalues

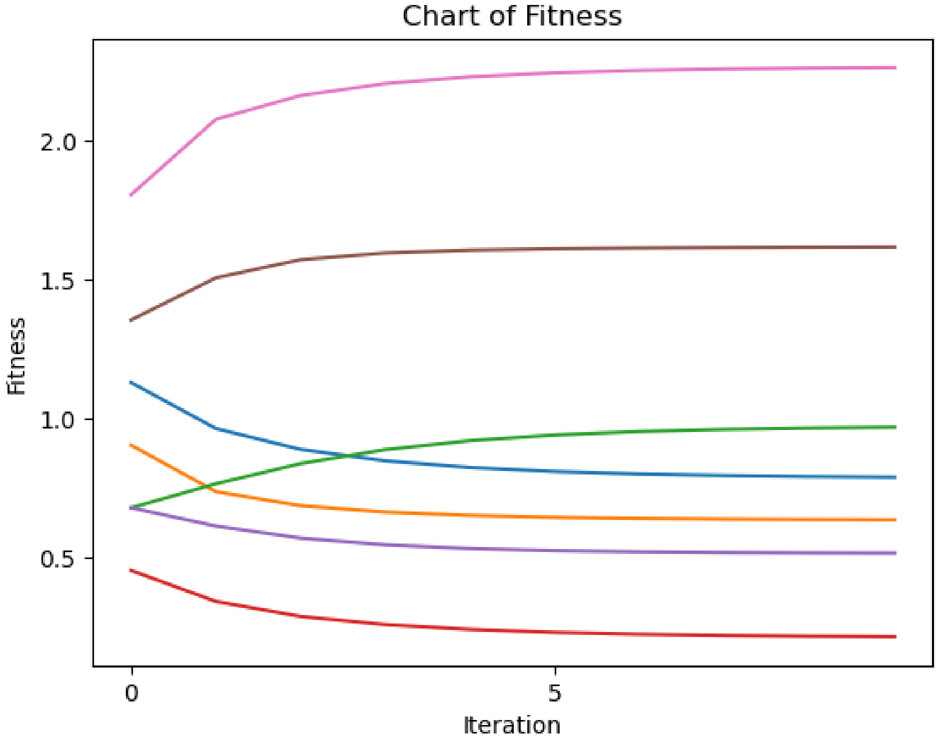
The ECI was calculated using the eigenvalues and countries were reordered from the most developed to the least

(for comparison are values with sign removed)

```
ECI as an eigenvalue problem:
         0: Vale of Arryn        = -0.56136
         1: The North            = -0.48151
         2: The Riverlands       = 2.0548
         3: Dorne                = -0.6958
         4: The Stormland        = -1.1651
         5: The Westerlands      = 0.30489
         6: The Reach            = 0.54408
---
Ordered:
        1. - (2) The Riverlands       = 2.0548
        2. - (6) The Reach            = 0.54408
        3. - (5) The Westerlands      = 0.30489
        4. - (1) The North            = -0.48151
        5. - (0) Vale of Arryn        = -0.56136
        6. - (3) Dorne                = -0.6958
        7. - (4) The Stormland        = -1.1651
Ordered (sign removed):
        1. - (2) The Riverlands       = 2.0548
        2. - (4) The Stormland        = 1.1651
        3. - (3) Dorne                = 0.6958
        4. - (0) Vale of Arryn        = 0.56136
        5. - (6) The Reach            = 0.54408
        6. - (1) The North            = 0.48151
        7. - (5) The Westerlands      = 0.30489
```

PART 2: Economic Complexity & Fitness Measures (2)

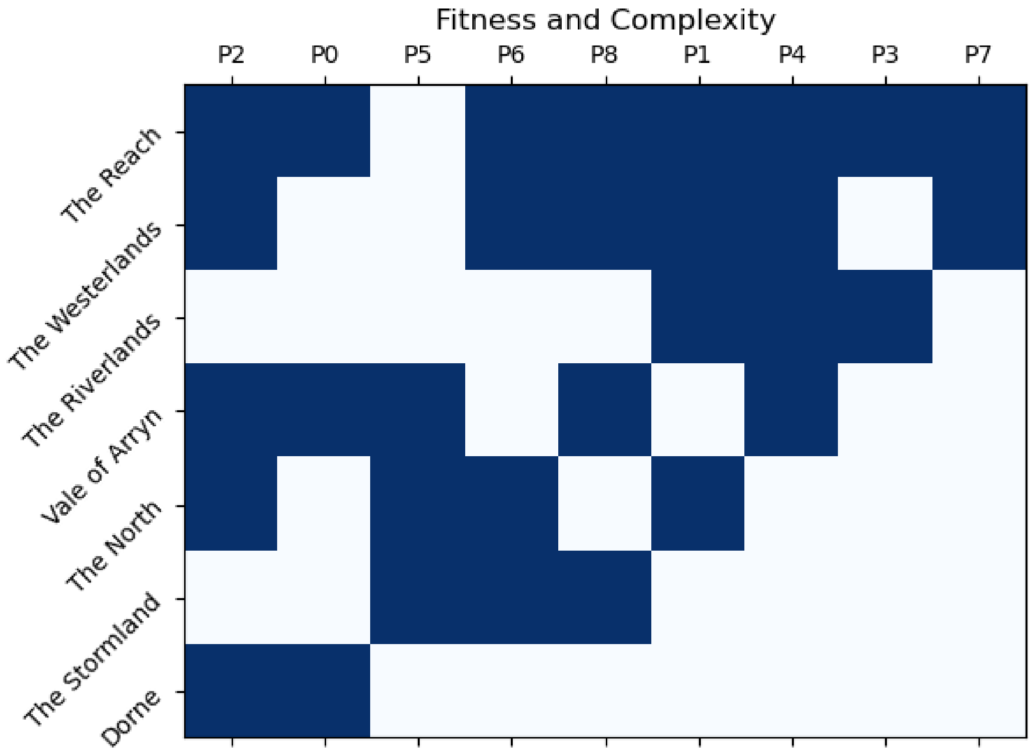




PART 2: Economic Complexity & Fitness Measures (3)

- ## Fitness and Complexity (1)

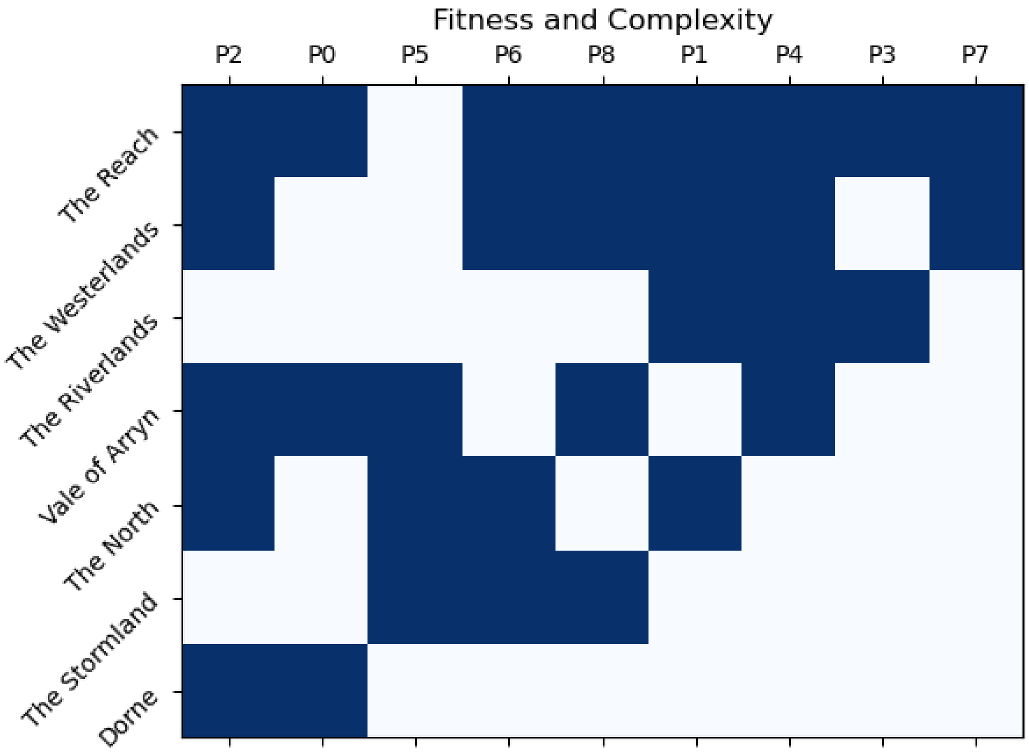
  This map shows that The Reach and The Westerlands are the most developed economies with the widest activities (production). Also we can see that the **P7** and **P3** are the most difficult products to make – as only two countries are able.



PART 2: Economic Complexity & Fitness Measures (4)

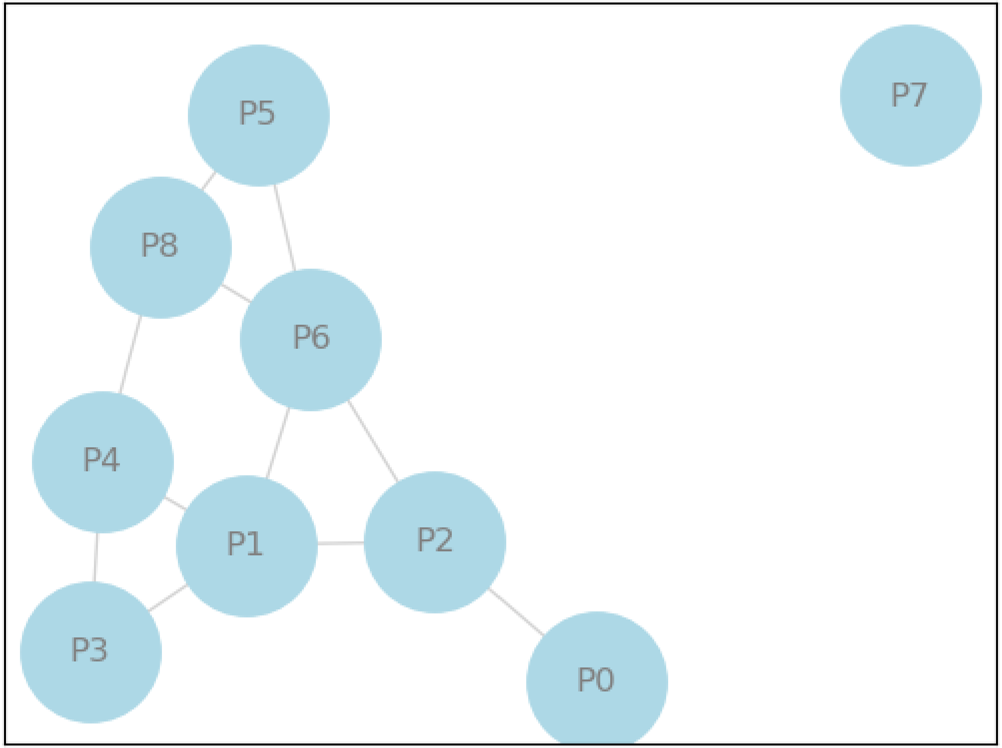
- **Fitness and Complexity (2)**

The Riverlands seems to specialize in top-end products of **P1**, **P4** and **P3** while country of Dorne is only capable of producing of the simplest of products **P2** (the most ubiquitous) and **P0**. Another specialized economy is The Stormland with only three products.



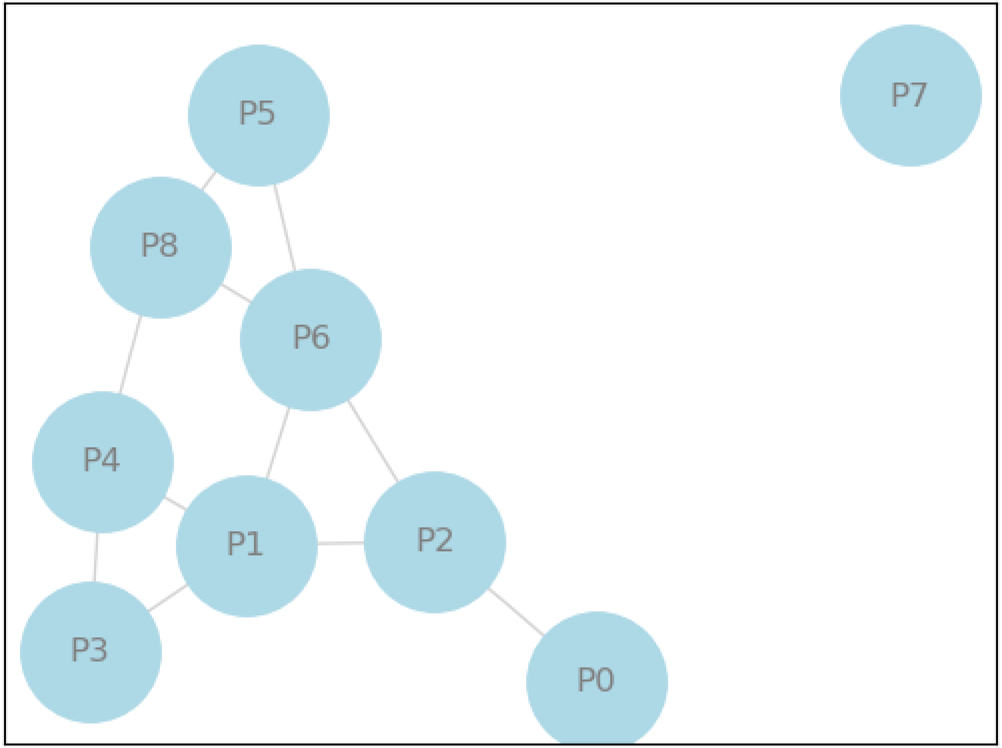
PART 3: Relatedness Between Activities (1)

- We can see that product **P7** is isolated from the rest of the network and it is the least ubiquitous product to make – **only two countries developed capabilities to produce it.**



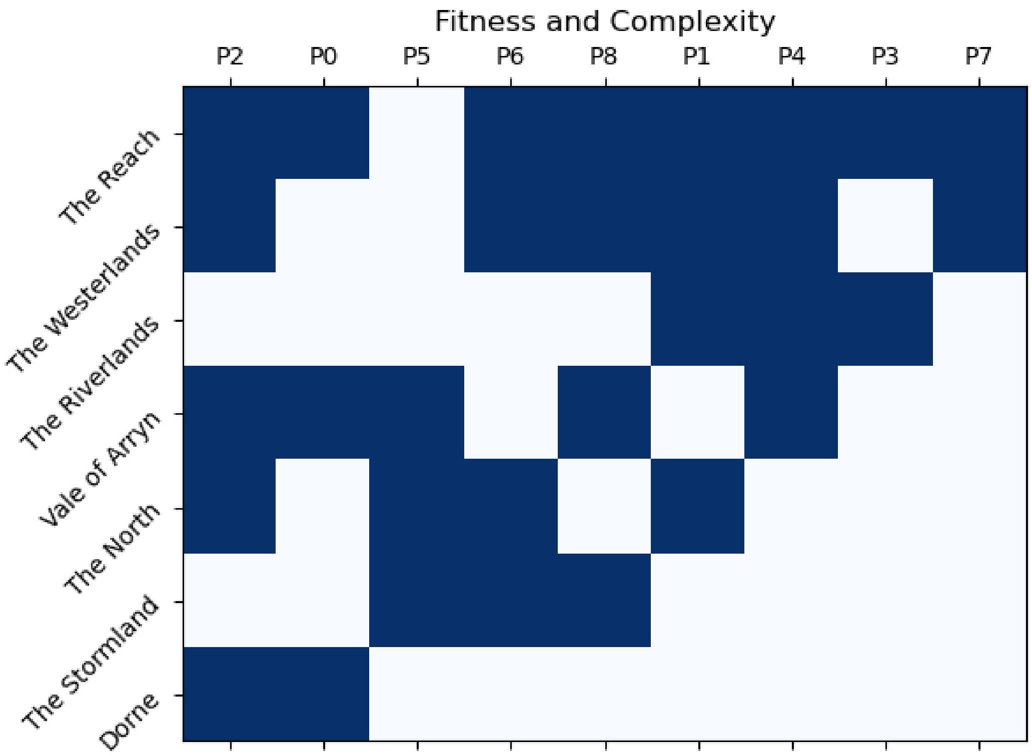
PART 3: Relatedness Between Activities (2)

- **P2** is the most ubiquitous product while **P1** and **P6** are entry points to the more wider range of activities. Every country having capability of producing P2 has capability of producing P0 (next slide).

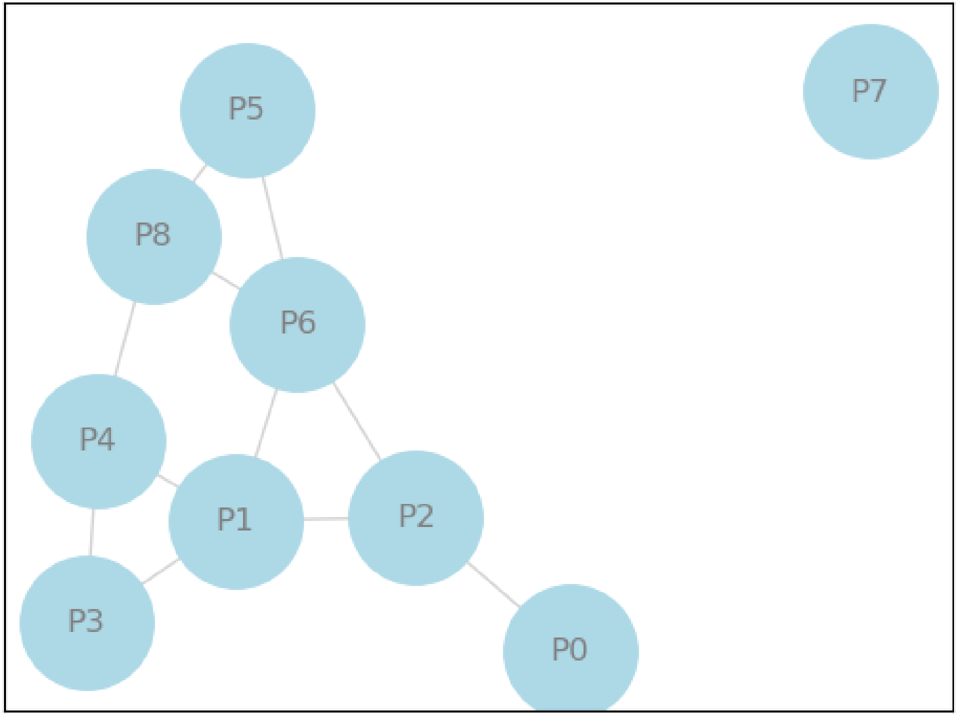


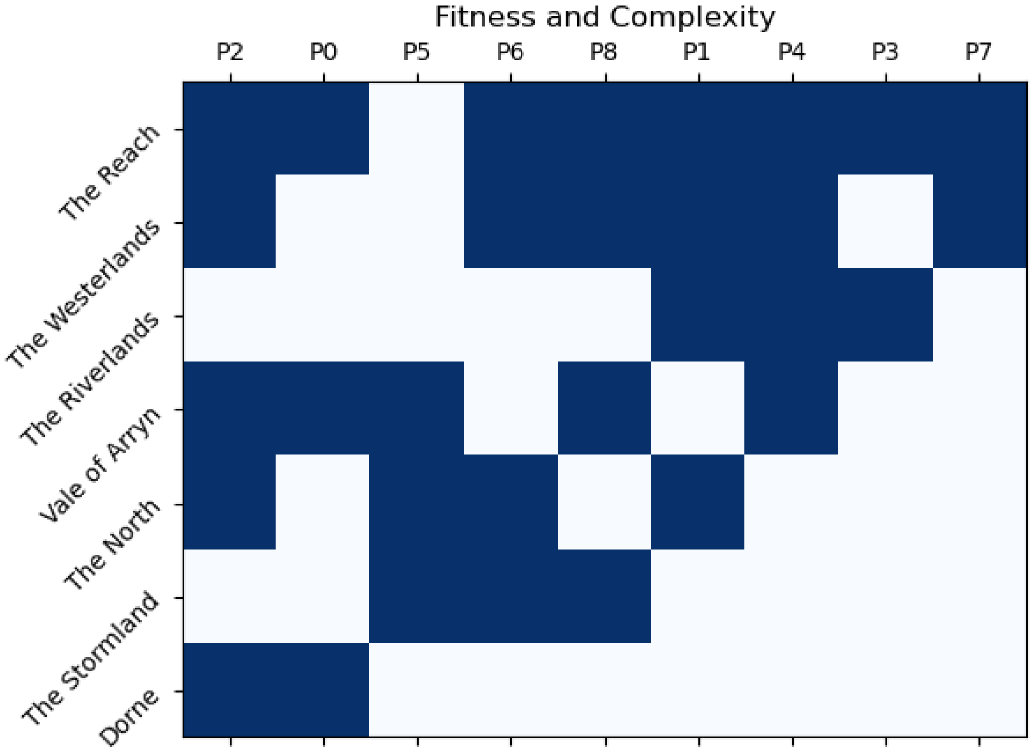
PART 3: Relatedness Between Activities (3)

- Here we can confirm that every country producing **P0** is also producing **P2**. We can also see that **P6** and **P1** splits the map in three segments: bottom not reaching P6 capability, top having P1 capability and mid between the two



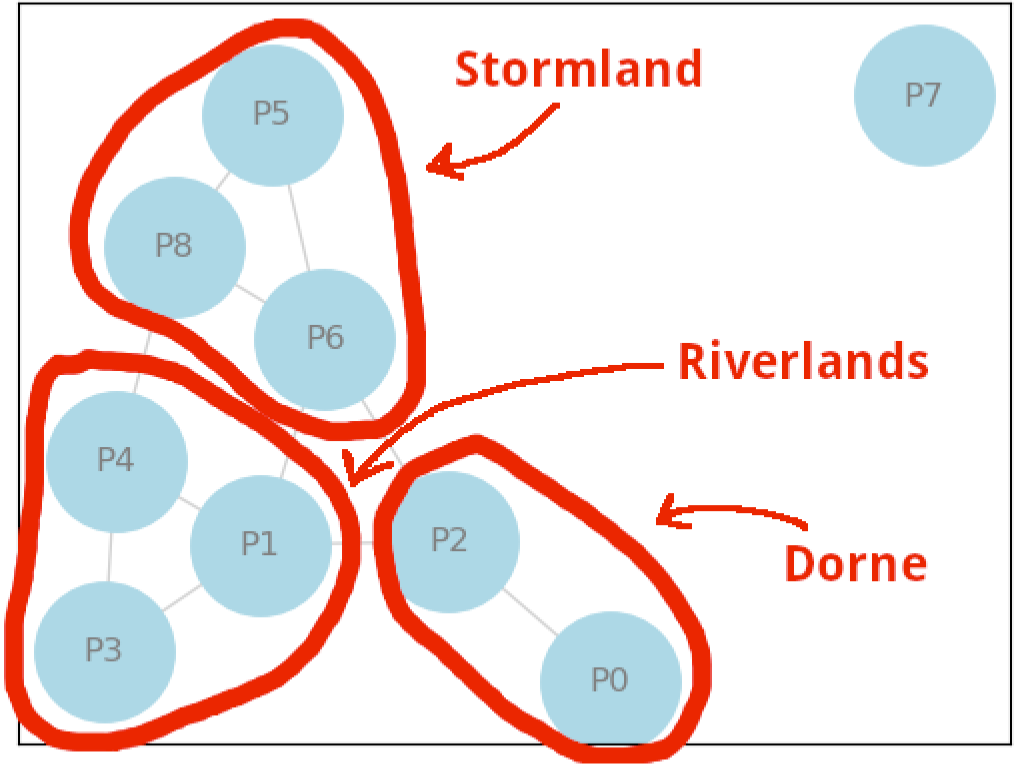
PART 4: Diversification Strategies (1)





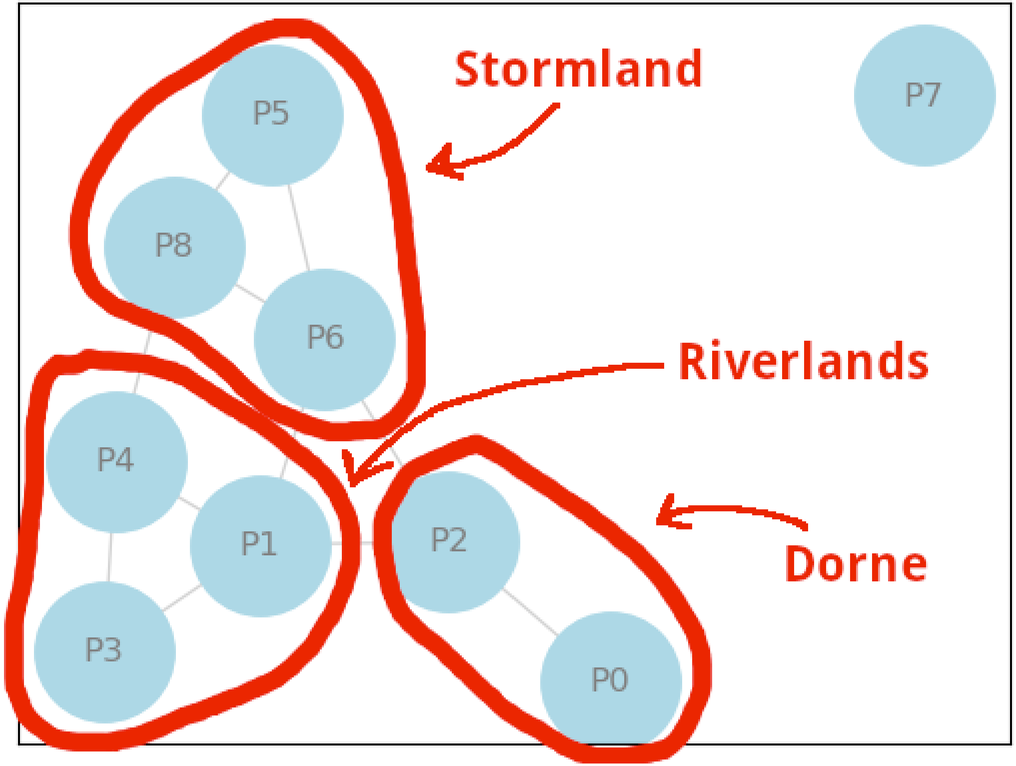
PART 4: Diversification Strategies (2)

- There is a reasonable path for **<u>Dorne</u>** to diversify their economy by trying to develop capability to produce product **<u>P6</u>** as that might open door for even more activities.



PART 4: Diversification Strategies (3)

- Similarly developing capability **P6** might help **The Riverlands** to access the more ubiquitous products of the lower segments. **Stormland** could on the other hand reach less ubiquitous products by trying to develop **P1** capability.



PART 5: Workflow and Contributions

- This slides and Jupyter (python) notebook is hosted on github:
  - https://github.com/ospalax/unu_merit_eci
- The collaboration started via initial email and then moved to a google's hangout chat room for offline async communication:
  - https://chat.google.com/app/chat/AAQAdfTIzyY
  - (invitation only)

Thank You